

Liquid lenses for aerospace beam steering and communications: MOSAIC

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Abstract: Laser communications (lasercom) can enable more efficient and higher bandwidth communications than conventional radio frequency (RF) systems, but requires more sophisticated pointing and tracking (PAT) systems to acquire and maintain links. Liquid lens arrays can provide compact, nonmechanical beam steering as an alternative to fast-steering mirrors and mechanical gimbals. An array of two liquid lenses offset in perpendicular axes along with a third on-axis lens in the array are used for beam steering and divergence control, respectively. The Miniature Optical Steered Antenna for Intersatellite Communications (MOSAIC) project applies liquid lens technology to create a transceiver for laser communications on spacecraft to enable wide field-of-view communications. This work provides analytical models of beam steering in order to inform subsystem sizing, and uses simulation studies along with previous work on space environment evaluation of liquid lenses to produce representative link budgets for a liquid lens based lasercom transceiver. A 25 Mbps link with 4 W transmit power at 1550 nm (optical C band) and 16-ary pulse position modulation (16-PPM) can be maintained up to 175 km separation with 3 dB margin, using larger pressure-actuated liquid lenses from Optotune arranged for hemispherical steering, potentially allowing for high speed optical links between formation flying swarms for applications such as interferometry.

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1. Introduction

Liquid lenses are compact, nonmechanically actuated focus-tunable lenses that use a liquid to change their focal length. Their low size, weight, power, and cost (SWaP-C) makes them attractive for aerospace applications, where they could be used for beam steering and communications. These properties make liquid lenses a potential candidate for use in satellite lasercom terminals, where they could be used to create lasercom terminals that can transmit and receive from a wide field of view, utilizing optical beam steering with liquid lenses.

This work is part of development for the Miniature Optical Steered Antenna for Intersatellite Communication (MOSAIC) project which aims to utilize liquid lenses for a hemispherically steering lasercom terminal for small satellites [1–5]. The MOSAIC project seeks to construct a compact, nonmechanical lasercom transceiver with integrated beam steering for small satellites using liquid lenses. This work formulates analytic models for beam steering based on ray transfer matrix extensions, validates analytic formulations with simulation results, and discusses constraints for sizing a lasercom transceiver on a volume-constrained satellite platform, as well as performing link budget analysis to analyze communications performance.

2. Background

Beam steering approaches in aerospace have unique challenges in balancing capabilities with size, weight, power, and cost. While mechanical gimbals are fully capable, their size and weight can make them unsuitable for example on small satellite platforms. Novel non-mechanical beam-steering solutions have therefore become popular, although have their own unique drawbacks. MEMS fast-steering mirrors (FSMs) are lightweight and low power, but have limited actuation

range and power handling capabilities [6,7]. Solutions such as fiber arrays and MOEMS coupled devices allow for integrated high speed beam steering on a discrete set of angles [8,9]. Liquid crystals can additionally be used for beam steering, although can have issues with power handling performance [10–12]. Liquid lenses have the potential to fill a gap in compact, non-mechanical beam steering with large apertures and improved power handling capability.

Liquid lenses have found applications in machine vision [13,14], phone cameras [15], microscopy [16], optical communications [17–21], and more due to their compactness and low power. There are two main techniques used to generate variable focusing power in liquid lenses: electrowetting and pressure-based. Electrowetting involves using an applied electric field causing an oil droplet's contact angle with an insulator to change [13], while pressure-based lenses utilize a voice coil or similar actuator, causing fluid to displace into a membrane [14].

Optotune and Corning Varioptic are the two main manufacturers of liquid lenses, each producing lenses with different operating principles. Corning Varioptic's liquid lenses employ electrowetting technology, whereas Optotune's lenses function based on pressure.

The transceiver design is based on a previous design by Zohrabi [22] utilizing a single on-axis lens for divergence control and an additional two lenses offset in x and y for 2D steering, initially proposed for LIDAR. A diagram of the steering system used for analysis in this work is shown in Fig. 1 with all associated dimensions annotated. The lens models used in this work include Optotune EL-16-40-TC lenses with ± 10 D focusing range and a 16.0 mm clear aperture, and Corning Varioptic A-39 lenses with $+15/-5$ D focusing power and 3.9 mm clear aperture.

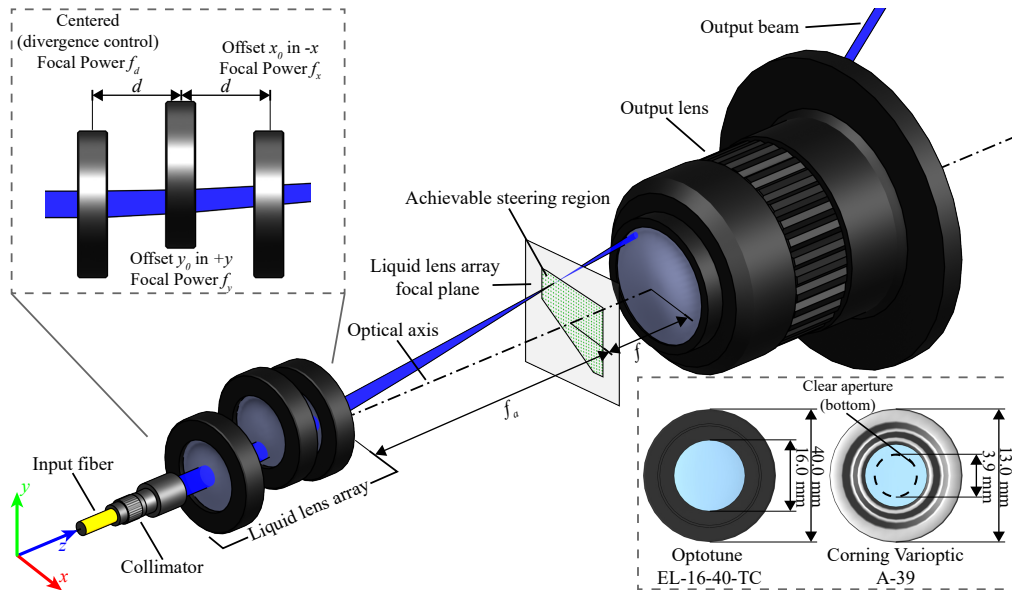


Fig. 1. Overview of liquid lens beam steering system, showing input fiber, liquid lens array used for beam steering, and output lens. The beam from the collimator goes through the liquid lens array, where the offset lenses steer the beam and the on-axis lens focuses the beam onto the focal plane, after which the output lens magnifies the angular displacement. Liquid lens device and aperture dimensions from Corning Varioptic and Optotune also pictured.

Previous work on space environment evaluation has indicated that liquid lenses can transmit high power continuous-wave lasers up to 2.25 W without measurable damage, and with temperature rises well within specification [4,5], significantly higher than the approximately 0.2 W that currently available commercial MEMS FSMs can operate with [6], although custom designed hardware could potentially handle more power.

Link budgets [3], steering models [4,22], and space environment evaluation [4,5] have also been studied separately. This work analyzes the transmitter, presenting analytical formulations of transceiver capabilities given space constraints, and also combines simulation studies with previous work on high power laser testing [4] and microgravity testing [5] into a representative link budget for an example liquid lens based lasercom terminal.

3. Analytical model

Modeling the system analytically helps to look at overarching trends without having to do computationally expensive simulation sweeps. In addition, analytic modeling is used to understand system constraints and limitations, allowing the overall transmit architecture to be sized to fit constraints, both in terms of physical space and steering range. Analytic models can additionally be integrated into steering and control algorithms as a feedforward controller, with additional active control for fine pointing.

The modeled optical system consists of the input beam only up to the focal plane of the liquid lens array, omitting the final lens used to amplify steering as shown in Fig. 1. The system consists of collimated input sequentially going into a set of three liquid lenses separated by a distance d and focusing inputs f_d , f_y , and f_x , with the first lens on-axis and the last two offset from the optical axis in y and x . This is followed by a distance f_a between the liquid lens array and a focal plane. The focal plane is necessary in this system as it allows collimated output, minimizing divergence and maximizing link range.

Since the overall steering range of the liquid lenses considered is relatively small, paraxial approximations for ray propagation can be made without excessive error in final output. The final steering output can still be estimated using a simple magnification relation with the output lens.

The main tool for modeling optical systems analytically is using ray transfer matrices, which can be used to trace rays through the system. Ray transfer matrix formulations exist for common optical elements [23] and can be combined to create a system matrix. However, standard formulations of ray transfer matrices are limited to two dimensions, and optical elements only along the optical axis, making them unable to model the presented transceiver design. Ray transfer matrices in homogeneous coordinates are developed in order to incorporate optical elements offset from the optical axis.

3.1. Ray transfer matrix extensions

Extensions to ray transfer matrices can be made to make them suitable for modeling 3D systems and to incorporate off-axis optical elements. Extending ABCD matrices to three spatial dimensions can be done simply by adding linearly independent dimensions for the y -axis, making sure to keep entries such as distances and focal lengths the same. Lifting the system into homogeneous coordinates by adding an additional constant element of 1 in $\mathbf{x}$ enables the use of translation matrices to offset the optical axis to other elements. The input vector for a ray then becomes the five element vector

$$\mathbf{x} = \begin{bmatrix} x \\ \theta_x \\ y \\ \theta_y \\ 1 \end{bmatrix}, \quad (1)$$

where x and y are the initial coordinates of the ray, and θ_x and θ_y are the initial angles of the ray for x and y respectively. Common optical elements can now be represented in this new formulation by stacking them diagonally for elements that do not have astigmatism.

3.2. Full system

Combining these extended elements then gives the ray transfer matrix for the entire system $\mathcal{M}_{\text{sys}}$ as given in [4] with a full elementwise breakdown. Considering only collimated inputs allows simplifications to be made to the system. Applying the imaging condition and forcing collimated inputs to come to a focal point, the condition $\mathcal{M}_{11} = 0$ [23] can be applied to obtain the required divergence controlling power f_d given inputs f_x and f_y . Limiting input rays to on-axis rays gives the location of focused image spots at the focal plane of the system as:

$$\mathbf{x}_{\text{out}} = \begin{bmatrix} \frac{f_a x_0}{f_x} \\ \frac{x_0}{f_x} \\ \frac{y_0 \left(f_a - d \left(\frac{f_a}{f_x} - 1 \right) \right)}{f_y} \\ - \frac{y_0 \left(\frac{d}{f_x} - 1 \right)}{f_y} \\ 1 \end{bmatrix}, \quad (2)$$

with the corresponding required input from the divergence controlling lens f_d given by

$$f_d = \frac{(f_x f_y - 2df_y - df_x + d^2) f_a + 2df_x f_y - d^2 f_x}{(d - f_x - f_y) f_a + f_x f_y - df_x}. \quad (3)$$

3.3. Steering constraints

Constraints on the focusing power of each lens can be applied to understand to which areas on the focal plane the liquid lens array can steer to. Each lens provides two constraints on maximum positive and maximum negative focusing power $f_{\text{max}+}$ and $f_{\text{max}-}$, yielding a six-sided constraint surface. Manipulating the results in Eq. (2) and Eq. (3) and substituting quantities of f_d , f_x , and f_y for maximally constrained values $f_{\text{max}+}$ and $f_{\text{max}-}$ gives rise to the following set of six constraining lines on the focal plane of the system:

$$f_d = f_{\text{max}+} : y = \left(-\frac{y_0 (2d - f_{\text{max}+})}{x_0 (d - f_{\text{max}+})} \right) x + \frac{y_0 (2dx_0 + f_a x_0 - f_{\text{max}+} \cdot x_0)}{x_0 (d - f_{\text{max}+})} \quad (4)$$

$$f_d = f_{\text{max}-} : y = \left(-\frac{y_0 (2d - f_{\text{max}-})}{x_0 (d - f_{\text{max}-})} \right) x + \frac{y_0 (2dx_0 + f_a x_0 - f_{\text{max}-} \cdot x_0)}{x_0 (d - f_{\text{max}-})} \quad (5)$$

$$f_x = f_{\text{max}+} : x = \frac{f_a x_0}{f_{\text{max}+}} \quad (6)$$

$$f_x = f_{\text{max}-} : x = \frac{f_a x_0}{f_{\text{max}-}} \quad (7)$$

$$f_y = f_{\text{max}+} : y = \frac{y_0}{f_{\text{max}+}} \left(f_a - d \left(\frac{x}{x_0} - 1 \right) \right) \quad (8)$$

$$f_y = f_{\text{max}-} : y = \frac{y_0}{f_{\text{max}-}} \left(f_a - d \left(\frac{x}{x_0} - 1 \right) \right) \quad (9)$$

From the reference constraint diagram in Fig. 2 and the analytic constraints given in Equations 3.8-3.13, some trends in the imaging plane can be observed. First, as previously observed in simulations [1,2], the two offset lenses act as a small Galilean telescope, where depending on lens command, both can work together to either contract or expand the beam. The same effect can be observed in Fig. 2, where in the upper right quadrant, the two off-axis lenses work to focus

the beam prematurely, which requires negative focusing power from the on-axis lens to focus on the focal plane. Similarly, the two off-axis lenses work together in the lower left quadrant to cause the beam to diverge, requiring additional positive focal power from the divergence control lens to focus.

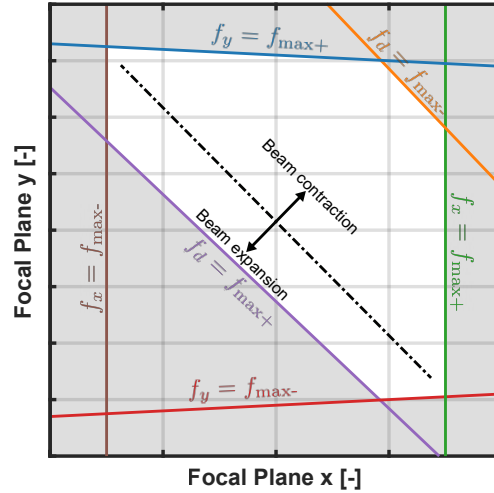


Fig. 2. Reference constraint space showing relative locations of all six constraints on focal plane. Units omitted since this is a reference and is not meant to represent any specific system.

Secondly, the impact of lens spacing d causes the constraining lines for f_y to be distorted away from parallel lines and into the sides of a trapezoid. This occurs due to the first on-axis lens causing the input beam to the second off-axis lens to be shifted, creating a coupled steering effect.

Steering constraints and sensitivity to system parameters can now be analyzed for specific cases. A reference system has been used as a starting point before modifying parameters. The reference parameters are given in Table 1.

Table 1. Default parameters of reference system.

Parameter	Symbol	Default Value	
		Corning Varioptic	Optotune
Lens spacing	d	5 mm	5 mm
Focal plane distance	f_d	100 mm	100 mm
X lens offset	x_0	1 mm	5 mm
Y lens offset	y_0	1 mm	5 mm
Maximum negative focal power	$f_{\max-}$	-5 D	-10 D
Maximum positive focal power	$f_{\max+}$	+15 D	+10 D

For a system with rotationally symmetric steering requirements, the main product from the constraints is the maximum steerable circle within the constraints, as shown in Fig. 3. Maximizing these quantities while minimizing the lens offsets to maximize clear aperture size can thereby be one way to optimize system parameters. Varying some of these parameters and looking at the maximum steerable radius can be considered a proxy for how suitable a given system configuration is for hemispherical steering.

Figure 4(a) shows the maximum image circle radius for varying lens distance d . The Corning Varioptic image circle remains a constant value as it is limited primarily by the $f_d = f_{\max+}$

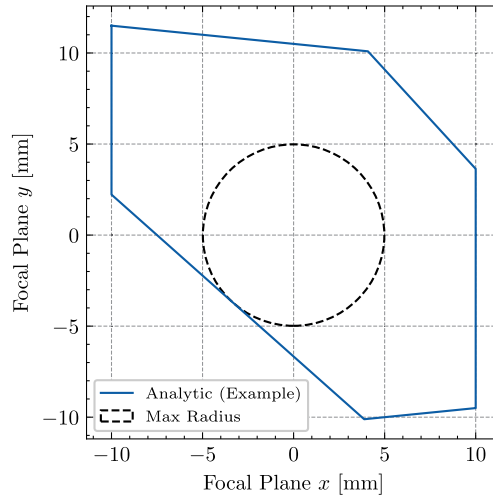


Fig. 3. Diagram showing maximum steerable circle from the origin governed by the steering constraints from example system.

constraint. On the other hand, the Optotune steering circle increases rapidly, up to a point where another constraint then governs the maximum steerable circle.

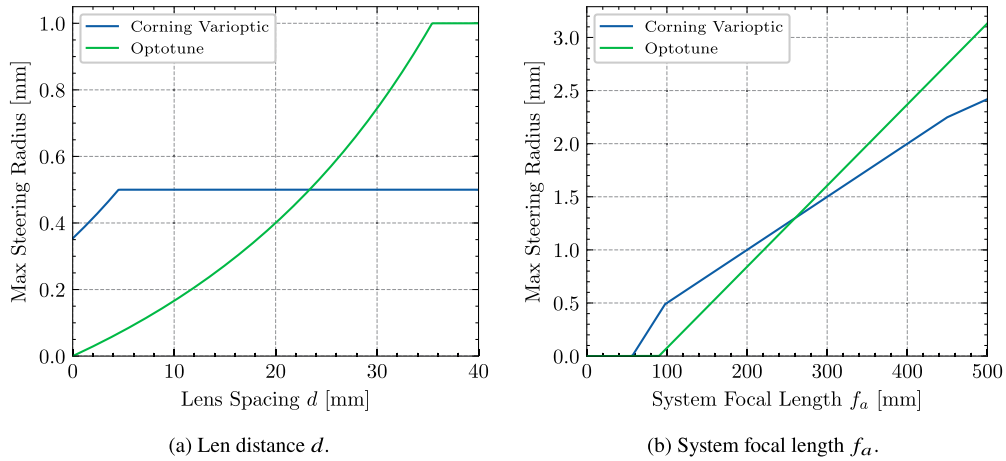


Fig. 4. Image plane limits for Corning Varioptic and Optotune lenses, for varying lens distance d and system focal length f_a .

Figure 4(b) shows the maximum steering circle for both Corning Varioptic and Optotune lenses for varying focal plane distance. Below a certain focal plane distance it can be seen that it is simply not possible at all to produce a focused spot on the focal plane; this is approximately at 46.6 mm for the Corning Varioptic lenses and 80.0 mm for the Optotune lenses in the reference configuration as shown in Table 1. Above this, the Optotune lenses linearly increase in steering circle radius, whereas the Corning Varioptic lenses hit different constraints, causing the piecewise linear shape seen in Fig. 4(b).

4. Simulation study

A Zemax model of the transmitter was constructed from the manufacturer-provided models of Corning Varioptic A-39N0 and Optotune EL-16-40-TC liquid lenses. Customization options for these lenses can limit the generalizability of single models. Here, the standard Zemax models, designed for visible light, were modified to accommodate shortwave infrared (SWIR) capabilities required in this work.

After representative models of the liquid lenses are constructed, a fisheye lens is added in series. The current system design places the fisheye lens 500 mm away from the liquid lens array, as shown in Table 2 so that the liquid lens' raw steering range can reach all points of the fisheye lens' input aperture. While this focal length is much greater than the minimum required as shown in Fig. 4, it is a useful comparison case between the lenses as both Corning Varioptic and Optotune lenses can obtain full hemispherical coverage with reasonable offsets in x and y that do not significantly limit the apertures of both lenses. An image of both Zemax models is shown in Fig. 6

Table 2. Optical parameters used in simulation for both the Corning Varioptic A-39N0 and Optotune EL-16-40-TC liquid lenses

Parameter	Symbol	Units	Corning Varioptic	Optotune
			A-39	EL-16-40-TC
Beam Width	ω_0	mm	1.6	3.0
Wavelength	λ	nm	1550	1550
Liquid lens offset	x_0, y_0	mm	1	5
Distance between LLs	d	mm	0.5	5
LL array focal length	f_a	mm	500	500
Fisheye focal Length	f_{fisheye}	mm	17.4	17.4

4.1. Comparison to analytic model

Figure 5 shows a comparison between the analytic model as presented in §3 and simulation studies using Zemax's physical optics propagation tool with parameters as given in Table 2 [1,4,24]. The Optotune model as in Fig. 5(b) fits the simulated data extremely well on the focal plane, with some of the simulated points being shifted towards the top-right. Initially, the Corning Varioptic analytic model as in Fig. 5(a) does not fit the simulated results well at all, however this is due to the full voltage range of the Corning Varioptic lenses actually resulting in a much larger range of focusing power. From the datasheet, fitting bounds of -16 D up to +29D [13] results in an extremely tight fit to the simulated points with similar top-right shift as observed in the Optotune model.

The top-right shift of the points implies that the negative focusing constraint of the on-axis lens is less tight than estimated, implying that there is a source of divergence in the simulated model not present in the analytic model. The source of this deviation is likely from the collimator being slightly more divergent than the planar input used for the analytic model.

The function linking commanded voltage/current to the focusing power is linear for the Optotune lenses which makes the points become more condensed towards the $x+$ limit in this linear sweep of focal command. However for the Corning Varioptic lenses, the focal command function is non-linear such that the points actually bunch up towards the bottom left of the focal plane, overriding the effect of the optical system bunching up the points towards the $x+$ constraint.

For a system designer, this comparison shows that the analytical model can be useful in sizing the initial system.

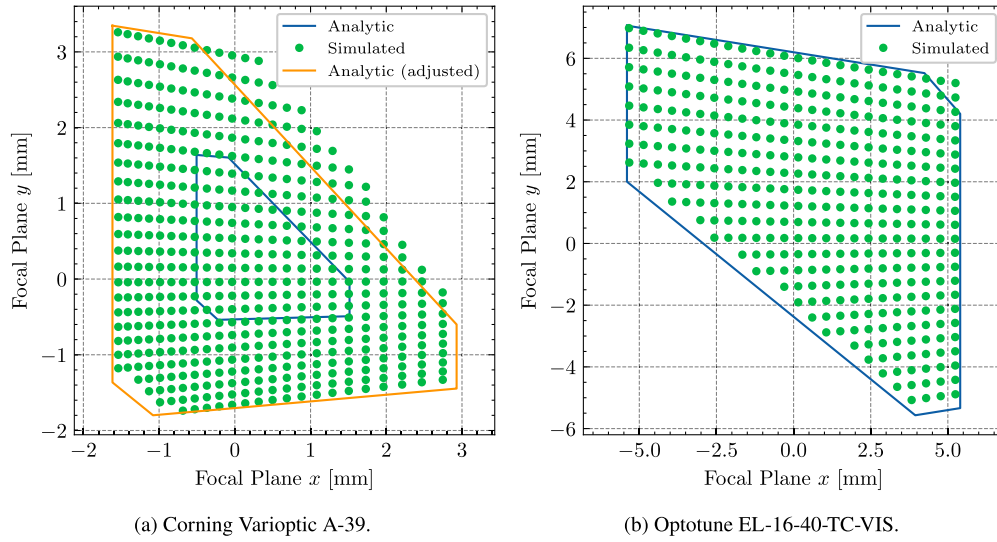


Fig. 5. Analytically derived image plane limits for Corning Varioptic A-39 and Optotune EL-16-40-TC lenses overlaid with simulation results from Zemax for an identical system as in Table 2. Lens commands are spaced 0.5 mA between -300 mA and 300 mA for Optotune data and 0.25 V between 0 V and 70 V for Corning Varioptic data.

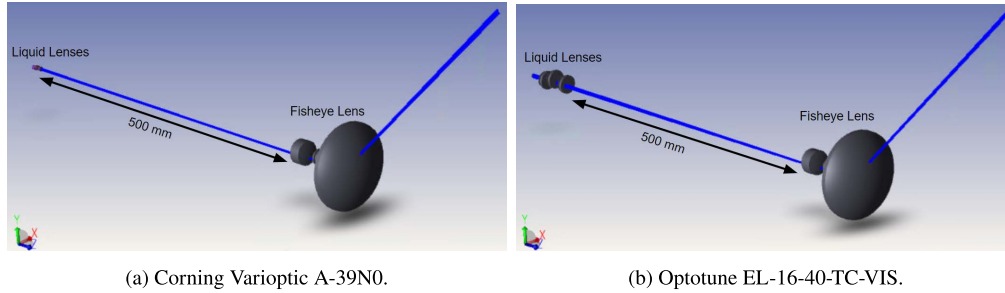


Fig. 6. Transmission optical trains modeled in Zemax with the Corning Varioptic A-39N0 and Optotune EL-16-40-TC-VIS lenses.

4.2. Transmit gain

The transmit gain is a measure of how directed a beam is compared to an isotropic radiator. In this system, the transmit gain represents how much more directed the beam exiting the fisheye lens is, in comparison to an isotropic radiator, incorporating all aspects of the transmitter. Transmit gain for an optical system can be calculated from the divergence, and is given as [25]

$$G_{\text{tx}} = \frac{8}{\theta^2} \quad (10)$$

where θ is the half-angle divergence of the beam. Conventionally, transmit gain is given in units of dBi, as gain over an isotropic radiator.

Transmit gain maps for Optotune and Corning Varioptic lenses in Fig. 8 and Fig. 9 obtained from the divergence maps in Fig. 7 show the same Galilean telescope effect, where in one quadrant where both work to reduce divergence, and an opposite quadrant where divergence is amplified. Prior results from 1D steering show a transmit gain and loss of ± 1 dB, an effect attributable to beam expansion. Optotune's results appear smoother as seen in Fig. 8, possibly due to the simple

curvature-based models, compared to the more complex Corning Varioptic models that consist of many ZPL scripts and could potentially represent the lenses more accurately.

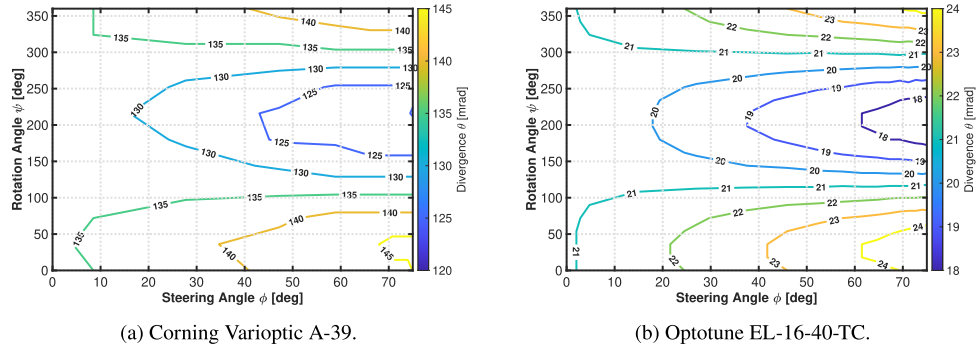


Fig. 7. Half-angle divergence measurements from simulated optical train for Corning Varioptic A-39N0 and Optotune EL-16-40-TC lenses.

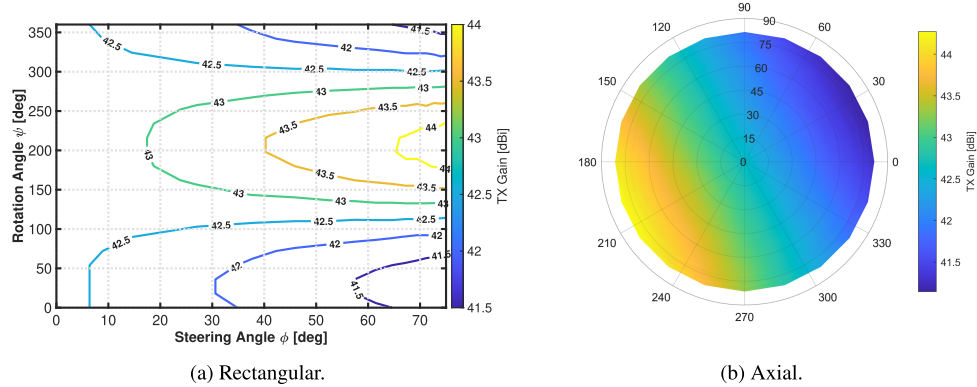


Fig. 8. Optotune transmit gain based on simulated divergence measurements across steering and rotation angles. Represented in both rectangular and axial projections.

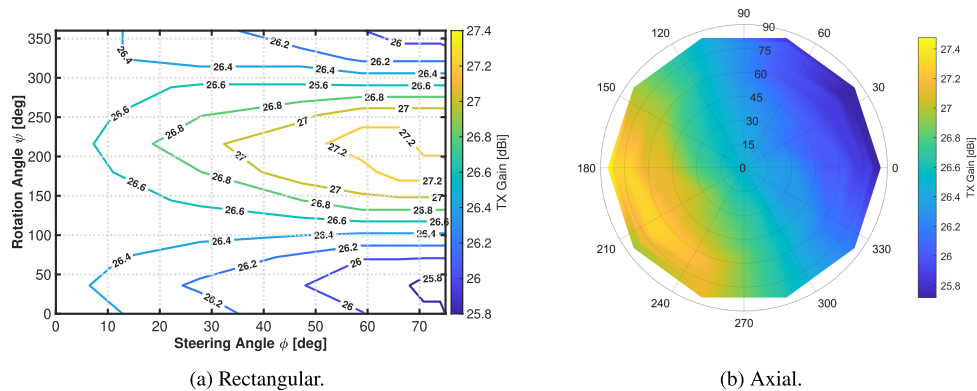


Fig. 9. Corning Varioptic transmit gain based on simulated divergence measurements across steering and rotation angles. Represented in both rectangular and axial projections.

The transmit gain in this system is significantly below contemporary optical communication systems can achieve. Modern free-space laser transmitters for space applications can reach 90+ dB transmit gain [7,26], but this system only accomplishes 42.8 dB and 26.5, dB for Optotune and Corning Varioptic lenses, respectively. This reduced gain can be attributed partially to the small apertures of the liquid lenses. Additionally, the significant angular magnifications necessary for hemispherical steering cause the beam to shrink, further increasing the divergence. However, since the transmit gain is much lower, pointing jitter can be discounted as a loss source, as the beam is essentially flat across conventional pointing errors in the 1-100 mrad range. The divergence of 17-100 mrad is also approximately the same magnitude or larger as the demonstrated pointing error for a similar terminal using liquid lens arrays of approximately 10-20 mrad [27].

4.3. Receive gain

The receive gain is directly proportional to the area of the collecting optics, given as

$$G_r = \frac{4\pi A_r(\phi)}{\lambda^2} \quad (11)$$

where $A_r(\phi)$ is the projected receiver area as a function of the steering angle ϕ . Multi-element lenses with internal aperture stops cannot be modeled analytically. Instead, they must be simulated to obtain a receive aperture curve or interpolated from values given in the datasheet provided by the manufacturer. Photographic lenses can limit the maximum receiving aperture as the goals of photography can be divergent from those of communication. In particular, ideal photographic lenses produce even illumination across their focal plane, whereas for communications, the ideal case is the maximum possible aperture at all receiving angles. The receiving aperture for a stop-limited system as a function of its relative illumination across the sensor diagonal $RI(x)$ can be given as

$$A_r(\phi) = \frac{\pi}{4} \left[\frac{f}{f/\#} \right]^2 \cdot RI \left(L_{\text{sensor}} \frac{2\phi}{\phi_{\text{max}}} \right) \quad (12)$$

where f is the focal length of the lens, $f/\#$ is the aperture stop of the lens, L_{sensor} is the length of the design image sensor's diagonal, ϕ is the receive incidence, and ϕ_{max} is the maximum field of view of the lens, which is doubled, as conventionally the field of view is listed as a full-angle field of view. If $RI(x)$ is not given, it can be quadratically interpolated from sensor-edge values. Receive gain can then be calculated as in Eq. (11).

5. Link budgets

In the baseline case, an avalanche photodiode (APD) is assumed to be the detector due to superior gain characteristics compared to PIN diodes and their technological maturity for operating 1550 nm [28]. Other receiving systems can use erbium doped fiber amplifiers (EDFAs) or semiconductor optical amplifiers (SOAs) for additional amplification, but both require fiber coupling, which brings additional loss. Instead, in this work we assume use of an APD.

Detectors from Voxel, Laser Components, and Hamamatsu are used as the baseline in this work [29–31]. The supporting electronic hardware consists of a transimpedance amplifier, which converts the avalanche photodiode's current responsivity into a voltage that can then be sampled by an analog to digital converter (ADC).

Signal to noise ratio is conventionally defined as the ratio of signal power to noise power [32], which is also given by the ratio of the square of the signal and noise currents, given by

$$\text{SNR} = \frac{P_{\text{signal}}}{P_{\text{noise}}} = \frac{i_{\text{signal}}^2}{i_{\text{noise}}^2} \quad (13)$$

Signal power can be modeled as

$$i_{\text{signal}} = RMP_{\text{signal}} \quad (14)$$

where M is the APD gain, and R is the responsivity of the APD. R can further be broken down into quantum efficiency if necessary, but in this case, the responsivity is given in Table 3. R in terms of quantum efficiency is

$$R = \frac{\eta q}{h\nu} \quad (15)$$

where η is the quantum efficiency, q is the fundamental charge, h is Planck's constant, and ν is the optical carrier frequency.

Table 3. Parameters of APD used, loosely based on Voxel, Hamamatsu, and Laser Components APDs [29–31].

Property	Symbol	Units	Value	Notes
Max gain	G	-	20.0	
Max bandwidth	B	MHz	275.0	CLICK filters to 200 MHz [7]
Responsivity	R	A/W	1.1	
Dark current	I_D	nA	25	
Noise equivalent power	P_{NEP}	dBm	-63.5	Over entire bandwidth

For noise sources which can be considered white, SNR can be approximated as:

$$\text{SNR} = \frac{i_{\text{signal}}^2}{\Sigma(n \cdot B)} \quad (16)$$

where n is the power spectral density of each individual noise contribution, and B is the receiving bandwidth.

5.1. Noise sources

Noise in an APD emerges from both optical and electronic effects. The noise sources considered in this link budget include shot noise, dark current noise, and thermal noise from the trans-impedance amplifier (TIA). In the context of an erbium-doped fiber amplifier (EDFA), we would consider amplified spontaneous emission (ASE) noise as well, but as this work uses an APD, ASE is not incorporated. While we can use the noise equivalent power (NEP) listed on datasheets for a preliminary comparison of different APDs, the noise in a link should not be treated as a constant. Given that shot noise varies with signal power and gain, and the type of electronics used can substantially impact the overall noise, a comprehensive noise model necessitates the consideration of all these factors.

Optical shot noise arises from the discrete nature of photons and sets fundamental limits on the system's SNR. Optical communication systems tend to have optical amplifiers with gain set such that shot noise is the dominant noise source in the system, making it shot noise limited. In an APD, it can be expressed as

$$n_{\text{shot}} = 2qRP_{\text{signal}}M^2F \quad (17)$$

where M signifies the APD gain and F represents the APD excess noise factor [25,33].

Dark current noise, which arises even in the absence of any illumination, behaves similarly to electronic shot noise with the addition of a small leaking current from the photodetector. Its amplification in APDs is particularly significant due to the avalanche effect. We can dissect dark current into device surface currents I_{DS} and bulk surface currents I_{DB} , with the latter traversing

the gain regions and consequently being subjected to multiplication [34]. This relationship can be delineated as

$$I_D = I_{DS} + I_{DB}M \quad (18)$$

Furthermore, the dark noise power spectral density associated with these multiplied currents can be approximated as

$$n_{\text{dark}} = 2q \left(I_{DS} + I_{DB}M^2 \right) \approx 2qI_{DB}M^2 \quad (19)$$

with high gain scenarios presuming that all the dark current present is bulk dark current [34].

Thermal noise, alternatively known as Johnson-Nyquist noise is caused by thermal agitation of charge carriers [35], and can be formulated as:

$$n_{th} = 4kT/R_{TIA} \quad (20)$$

where k is Boltzmann's constant, T is the temperature of the resistor, and R_{TIA} is the resistance of the feedback resistor of the TIA as shown in Fig. 10.

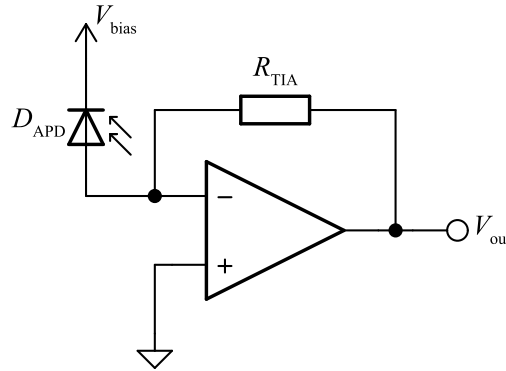


Fig. 10. Standard configuration of avalanche photodiode and transimpedance amplifier.

5.2. Modulation schemes

Direct detection with M-ary Pulse Position Modulation (PPM) is considered as a baseline for the modulation scheme in this work as it is simple, requiring the least amount of supporting hardware, and is well suited to average-power limited amplifiers such as Erbium Doped Fiber Amplifier (EDFA) [25,36] based transmit systems. Fiber-coupling is another option, [37], but requires more in-depth analysis, and direct detection is preferred in comparable mission architectures [7,26,38].

Since the receiver architecture has not been fully defined and finalized, we consider an example link with PPM characteristics, although many communication protocols would be suitable depending on the choice of detector and electronics, including coherent techniques if additional sensitivity is required.

A requirement on bit error probability P_b , based on a Poisson optical channel is used to determine the required bit energy to noise spectral density E_b/N_{0r} , from which a carrier to noise ratio C/N can be prescribed based on available bandwidth and desired data rate. For the simplest case of on-off keying, where each pulse carries a single bit of information, the bit error rate for an ideal photon-counting detector can be approximated for large E_b/N_0 as: [36]

$$P_{b(\text{OOK})} \approx Q \left(\sqrt{E_b/N_0} \right) \approx \frac{1}{2} e^{-\frac{1}{2} E_b/N_0} \quad (21)$$

where $Q(z)$ is the probability that a normally distributed random variable will return a value greater than z standard deviation above the mean, which is related to the complement of the error

function $\text{Erfc}(z)$ as shown in Eq. (22) [36].

$$Q(z) = \frac{1}{\sqrt{2\pi}} \int_z^\infty e^{-x^2} dx = \frac{1}{2} \text{Erfc}\left(\frac{z}{\sqrt{2}}\right) = \frac{1}{2} \left(1 - \text{Erf}\left(\frac{z}{\sqrt{2}}\right)\right) \quad (22)$$

For an M-ary PPM signal where each slot carries $1/\log_2 M$ bits of information, the relation between P_b and E_b/N_0 can be approximated as shown in Eq. (23) [36]. This approximation is valid for small M and improves with larger received signal powers.

$$P_{b(\text{M-PPM})} \approx \frac{1}{2} e^{-\frac{E_b}{N_0} \log_2 M} \quad (23)$$

Links are analyzed using the conventional link budget equation, modeling received power using the transmit power and all relevant losses in the system. The receive power P_{rx} can therefore be given by all the gains multiplied with all the losses, given by:

$$P_{rx} = P_{tx} L_{tx} L_p G_{tx} L_{space} L_{rx} G_{rx} \quad (24)$$

where L_{tx} is the transmit line loss, L_p is the loss due to pointing error, G_{tx} is the transmit gain, L_{space} is the free space loss, G_{rx} is the receive gain, and L_{rx} is the receive line loss [25].

The free space loss L_{space} is given by the Friis equation as shown in Eq. (25) [25], where λ is the carrier wavelength set at 1550 nm, and r is the separation distance between the transmitter and receiver.

$$L_{space} = \left(\frac{\lambda}{4\pi r}\right)^2 \quad (25)$$

Assembling all the pieces from previous sections, the final link budgets with parameters in Table 4 and detector parameters in Table 3 are shown in Table 5 and Table 6 for Optotune and Corning Varioptic lenses respectively. A high input power of 2.25 W was chosen based on excellent power handling results as described in [4].

Table 4. Parameters for final link budget.

Parameter	Symbol	Units	Value	Notes
Slot/Pulse Rate	S	MHz	400.0	
Wavelength	λ	nm	1550.0	
PPM Order		-	16	As used on CLICK [7,26]
System temperature	T_{sys}	K	300.0	Baseline
Data rate	R_b	Mbps	25.0	
Bandwidth	B	MHz	100.0	$m/\log_2 m$ expansion [36]
Bit error rate	P_b	-	1×10^{-4}	
Relative velocity		m/s	0.0	Assuming no Doppler shift
Lens Focal Length		mm	18	
Lens Aperture		-	f/1.4	

5.3. Link ranges

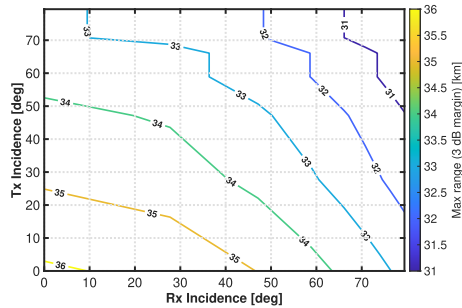
The range map based on transmit and receive incidence can be seen in Fig. 11. The maximum range is the range achievable at 3 dB margin with minimum transmit gain at a particular transmit and receive incidence. For on-axis links with this receive architecture, the Optotune lenses can close a 300 km link, dropping down to approximately 220 km at complete hemispherical steering. The Corning Varioptic lenses can close a 48 km link on-axis, dropping to 40 km at complete hemispherical steering, mainly due to the much higher divergence incurred by using such a small beam.

Table 5. Link budget for Optotune EL-16-40-TC lenses in described receiver architecture, for mutually on-axis and mutually 75° incidence cases using link parameters in Table 4 and Table 3.

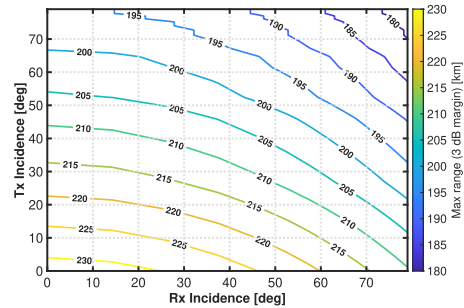
Parameter	Symbol	Units	On-axis	75 deg	Notes
Tx Power	P_{tx}	dBm	33.5	33.5	2.25 W
Tx Gain	G_{tx}	dBi	42.6	41.1	
Tx Line Loss	L_{tx}	dB	-1.5	-1.5	Allocated
Pointing Loss	L_p	dB	-1.0	-1.0	Allocated
Free Space Loss	L_{space}	dB	-243.0	-243.0	175 km range
Rx Gain	G_{rx}	dBi	88.3	87.6	18 mm, f/1.4
Rx Line Loss	L_{rx}	dB	-3.5	-3.5	
Rx Power	P_{rx}	dBm	-84.6	-86.8	
Signal to noise power ratio	C/N	dB	5.4	1.2	
Bit energy to noise density	E_b/N_0	dB	11.4	7.2	$C/N \cdot B/R_b$
Required E_b/N_0	E_b/N_{0r}	dB	3.3	3.3	
Margin		dB	8.1	3.9	

Table 6. Link budget for Corning Varioptic A-39 lenses in described receiver architecture, for mutually on-axis and mutually 75° incidence cases using link parameters in Table 4 and Table 3

Parameter	Symbol	Units	On-axis	75 deg	Notes
Tx Power	P_{tx}	dBm	33.5	33.5	2.25 W
Tx Gain	G_{tx}	dBi	26.5	25.7	
Tx Line Loss	L_{tx}	dB	-1.5	-1.5	Allocated
Pointing Loss	L_p	dB	-1.0	-1.0	Allocated
Free Space Loss	L_{space}	dB	-227.7	-227.7	30 km range
Rx Gain	G_{rx}	dBi	88.3	87.6	18 mm, f/1.4
Rx Line Loss	L_{rx}	dB	-3.5	-3.5	
Rx Power	P_{rx}	dBm	-85.4	-86.9	
Signal to noise power ratio	C/N	dB	3.4	0.5	
Bit energy to noise density	E_b/N_0	dB	9.4	6.5	$C/N \cdot B/R_b$
Required E_b/N_0	E_b/N_{0r}	dB	3.3	3.3	
Margin		dB	6.1	3.2	



(a) Corning Varioptic A-39.



(b) Optotune EL-16-40-TC.

Fig. 11. Minimum link ranges for Corning and Optotune lenses for varying transmit and receive incidence angles, with parameters in Table 3 and Table 4.

6. Transceiver architecture

The baseline transceiver design is a direct-detection system, as shown in Fig. 12. The transmit path uses a beam splitter to send a portion of the beam to a camera for active pointing control, with the rest passing through a fisheye lens. The receive path enters through the fisheye lens, using the same beam splitter to direct the signal to a mirror, liquid lens array, feedback camera, and detector. In order to balance the transmit and receive paths, a 50/50 beam splitter must be used.

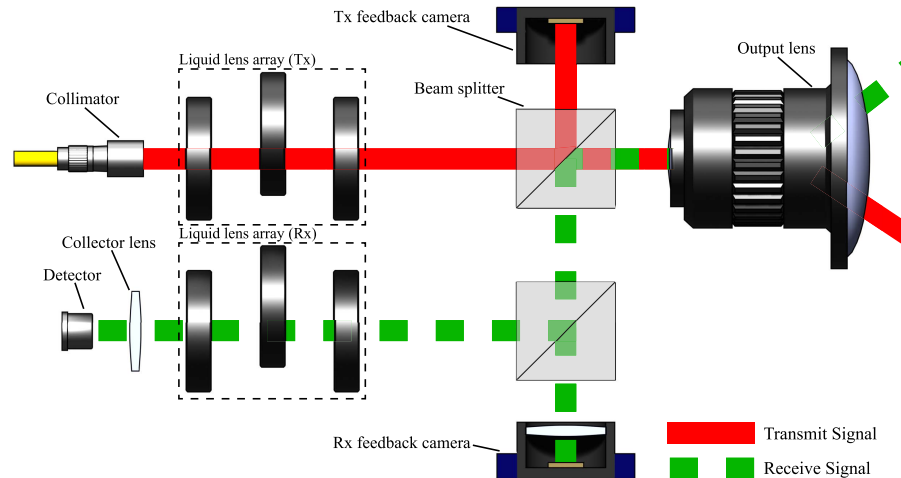


Fig. 12. Baseline transceiver design, showing the required peripherals and receive and transmit paths separated by the beam splitter [1].

Alternatively, a polarizing beam splitter (PBS) can be used with a wave plate such that links are circularly polarized, as shown in Fig. 13. Incoming beams are converted back to linear polarization, allowing separation without loss, enabling longer link ranges or higher data rates than the 3 dB loss intrinsic to the 50/50 beam splitter approach. Additionally, light leakage could affect receive camera sensitivity with 50/50 beam splitters, but this can be mitigated in the case of circularly polarized beam splitters, and further mitigated with filters if the transmitter and receiver are on slightly different wavelengths. However, as the beams going into the wave plate are not collimated or perfectly on-axis, some loss will still be incurred as the circular polarization will not be perfect. This configuration is shown in Fig. 11(b).

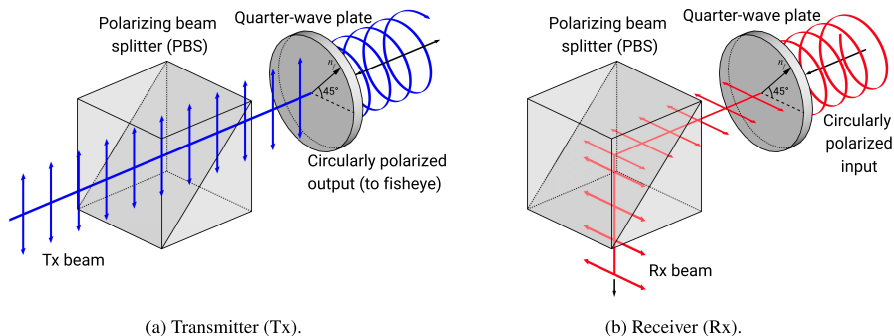


Fig. 13. Diagram of how polarizing beam splitters and quarter-wave plates can be used as to separate transmit and receive paths, similar to three-port optical circulators.

For a fully capable transceiver, the receiver must accept incoming signals from the same angles that the transmitter can steer towards. Since the liquid lenses govern steering constraints in the system, the easiest way to achieve transmit/receive symmetry is to have a reversible system, such that the lens commands required to receive a signal from a particular angle are the same ones required to steer and transmit to that angle. Utilizing a reversible system needs an additional lens on the receive path to convert the infinite conjugate system into a finite conjugate, focusing the input beam onto the detector.

In the case of hemispherical steering, the same aperture is required for transmit and receive due to the fisheye lens potentially obstructing certain angles if there are separate receive and transmit apertures. However, even slightly limited receive and transmit angles allow for separation into separate apertures, eliminating issues with light leakage and loss due to beam splitters.

7. Conclusions and future work

Overall, liquid lenses are shown to be a promising technology for miniaturized nonmechanical beam steering and free-space optical communications. The analytic formulations presented in this work show a clear method for subsystem sizing to meet requirements. The link budget analysis in concert with previous work on space environment testing [4] shows that hemispherical links for small satellites using liquid lenses are possible and offer certain advantages over other beam steering technologies – specifically in aperture size and power handling capability.

Key tradeoffs, including optical train length, maximum range, weight, and volume requirements, need careful evaluation based on mission parameters and multidisciplinary design optimization. The most constraining factor is that higher angular magnification results in larger beam divergence, reducing overall range.

While the range efficiency of the liquid lens transceivers as presented in this work is lower than that of equivalent mechanically steered systems, the 30 km and 175 km link ranges obtained from the Corning Varioptic and Optotune lenses respectively are still useful for applications such as formation flying, distributed aperture systems, such as for interferometry missions. Additionally, fisheye lenses and hemispherical steering are the most extreme cases for using liquid lenses for lasercom beam steering – significantly higher link ranges and beam quality can be obtained when required steering angle is limited smaller than the full hemisphere range, with the same analysis presented in this work, with the same advantages in aperture size and power handling that liquid lenses provide.

Liquid lenses also offer potential for new integrated beaconing techniques. Future work includes exploring pulsed beacons with controlled divergence, low data rate beaconing links, and utilizing free-space nutation [39] or spiral search beaconing.

Future work also involves end-to-end integration testing of terminal prototypes to validate the simulation work presented. Active control of the lenses based on feedback from a camera should also be explored, considering the cross-coupled control of the liquid lens array. A feedforward control system can be easily derived from the analytic beam steering formulations presented, as a starting point.

Funding. National Aeronautics and Space Administration (80NSSC19K0217).

Acknowledgment. The authors would like to thank Dr. Sarah Tedder at NASA Glenn Research Center for her support and correspondence over the course of the program. Additionally, the authors appreciate the work of Graham Wilcox in helping with manuscript review.

Disclosures. The authors declare no conflicts of interest.

Data availability. Data underlying the results presented in this paper are not publicly available at this time but may be obtained from the authors upon reasonable request.

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